AERO50001 Aerodynamics 2 Problem Set #3

Tutorial 1

3.1 Rounded leading-edge aerofoil

★☆☆ 10-15 mins

A symmetrical aerofoil with a rounded leading edge is flying at $M = 3$ at zero incidence and at sea level conditions ($T_1 = 288$ K, $p_1 = 1.013 \cdot 10^5$ N/m^2).

Hint: a curved shock forms in front of the aerofoil, however just in front of the nose it is a normal shock.

(a) Calculate the temperature and pressure at the nose of the aerofoil.

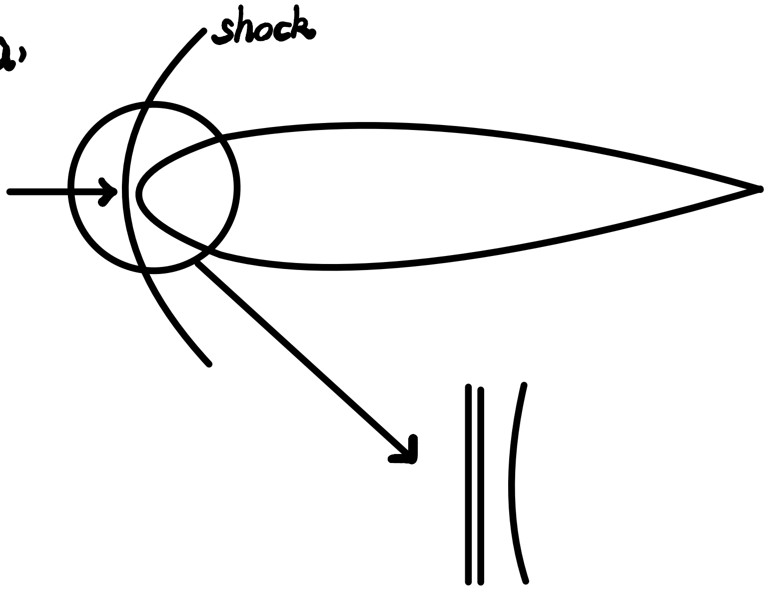
Hint: the nose of the aerofoil is a stagnation point.

$Note: to enter numbers in scientific format, use format 1.5e6, of instance]

(b) Determine the pressure coefficient at the nose of the aerofoil.

(c) What is the pressure coefficient at a point on the aerofoil where the local Mach number is equal to 1?

(a)



$M = 3$,

According to the table

$\frac{p_2}{p_1} = 10.33$, $\frac{\rho_2}{\rho_1} = 3.857$

$M_2 = 0.4752$

$\therefore p_2 = 10.33 p_1 = 1.046 \times 10^6$ Pa

$\because \frac{T_2}{T_1} = (\frac{p_2}{p_1})(\frac{\rho_1}{\rho_2})$

$\therefore T_2 = \frac{10.33}{3.857} T_1 = 771.3$K

$\therefore T_{02} = T_2 (1 + \frac{\gamma - 1}{2} M^2) = 806.17$K

$\therefore p_{02} = p_2 (1 + \frac{\gamma - 1}{2} M^2)^{\frac{\gamma}{\gamma - 1}} = 1221389$ Pa $= 1.22 \times 10^6$ Pa

(b) $C_p = \frac{p - p_\infty}{\frac{1}{2}\rho V_\infty^2} = \frac{p - p_\infty}{\frac{1}{2}\frac{p_\infty}{RT_\infty}(M_\infty\sqrt{\gamma R T_\infty})^2} = (\frac{p}{p_\infty} - 1)\frac{1}{\frac{1}{2}\gamma(M_\infty)^2}$

$$\therefore C_p \text{ at nose} = \left(\frac{p_{02}}{p_1} - 1\right) \frac{1}{\frac{1}{2}\gamma M_1^2} = 1.755$$

(c) $$C_p \text{ at aerofoil} = \left(\frac{p_3}{p_1} - 1\right) \frac{1}{\frac{1}{2}\gamma M_1^2}$$

$$= \left(\frac{p_3}{p_{02}} \frac{p_{02}}{p_1} - 1\right) \frac{1}{\frac{1}{2}\gamma M_1^2}$$

$$= 0.8523$$

3.2 Maximum mass flow rate

★★☆ 10-20 mins

Find an expression for the **maximum** mass flow rate $\dot{m}$ through a convergent-divergent nozzle as a function of γ, the throat area A_t, and the reservoir density and speed of sound, ρ_0 and a_0 respectively.

$Nǫte: for exponents, use ** instead of ^.]

$$\dot{m}^* = \rho^* u^* A_t = \rho^* M a^* A \xlongequal{M=1} \rho^* a^* A_t$$

$$= \rho_0 \left(1 + \frac{\gamma-1}{2}\right)^{\frac{-1}{\gamma-1}} a^* A_t = \rho_0 \left(\frac{2}{\gamma+1}\right)^{\frac{1}{\gamma-1}} a^* A_t$$

$$a^* = \sqrt{\gamma R T^*} = \sqrt{\gamma R T_0 \left(1 + \frac{\gamma-1}{2} M^2\right)^{-1}} \xlongequal{M=1} a_0 \left(\frac{2}{\gamma+1}\right)^{\frac{1}{2}}$$

$$\therefore \dot{m} = \rho_0 \left(\frac{2}{\gamma+1}\right)^{\frac{1}{\gamma-1}} a_0 \left(\frac{2}{\gamma+1}\right)^{\frac{1}{2}} A_t$$

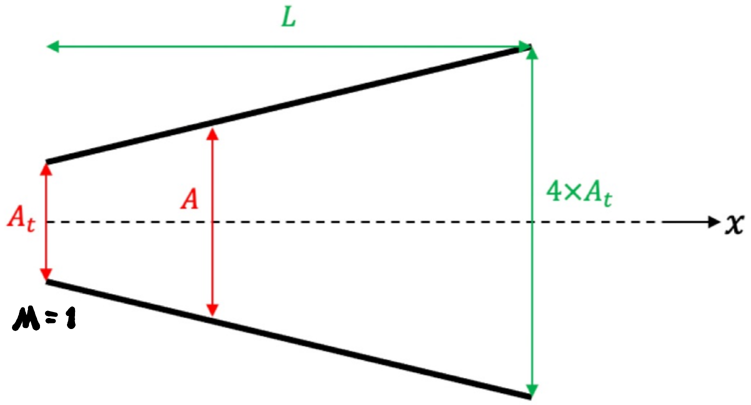
$$= \rho_0 a_0 A_t \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

3.3 Conical supersonic nozzles

★★☆ 15-25 mins

Consider a conical, supersonic nozzle, in which the area expands from A_t at the sonic throat to $4A_t$ at the exit, which is a distance L from the throat.

$Assume air with $\gamma = 1.4$].



(a) Find an expression for the diameter of any section along the nozzle D as a function of the diameter at the throat D_t and the non-dimensional distance from the throat x/L.

(b) Find an expression for the area of any section along the nozzle A as a function of the area at the throat A_t and the non-dimensional distance from the throat x/L.

(c) Using the expressions you have found, calculate the area and Mach number at a section located a distance $0.6L$ from the throat.

(d) OPTIONAL: plot (manually or using MATLAB) the variation of area and Mach number along the nozzle. You may assume any radius at the throat, i.e. 1.

(a) $A_t = \frac{\pi D_t^2}{4}$

$4 \times A_t = \frac{\pi D^2}{4} = \pi D_t^2$

$\therefore$ At the $x = L$, $D^2 = 4D_t^2$ $\quad \therefore D = 2D_t$

$\therefore D = \alpha \frac{x}{L} + \beta$

At $\frac{x}{L} = 0$, $D = D_t$ $\quad \therefore \beta = D_t$

At $\frac{x}{L} = 1$, $D = 2D_t$ $\quad \therefore \alpha = D_t$

$\therefore D = D_t \frac{x}{L} + D_t = D_t(\frac{x}{L} + 1)$

(b) $A = \frac{\pi D^2}{4} = \frac{\pi [D_t(\frac{x}{L}+1)]^2}{4} = \frac{\pi D_t^2}{4}(\frac{x}{L}+1)^2 = A_t(\frac{x}{L}+1)^2$

(c) At $\frac{x}{L} = 0.6$, $A = 2.56 A_t$

$\therefore \frac{A}{A_t} = 2.56$

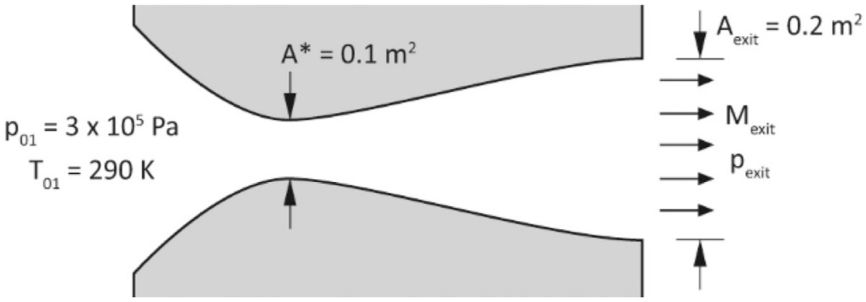
$\therefore M = 2.45$

(e) ...

3.4 Convergent-divergent nozzles

★★★ 20-30 mins

The figure below shows a convergent-divergent nozzle. It is observed that the flow (air, $\gamma = 1.4$) exits the nozzle as a perfectly expanded (isentropic) supersonic flow.



(a) Determine the mass flow rate through the nozzle.

(b) Determine the Mach number at the nozzle exit (M_{exit})

(c) Determine the pressure at the nozzle exit (p_{exit}).

(d) The pressure at the nozzle exit is gradually increased from its original value. For what range of values of p_{exit} would the mass flow rate through the nozzle remain unchanged?

(a) $$\dot{m}^* = \rho^* u^* A_t \xrightarrow{M=1} \frac{p^*}{RT^*}\sqrt{\gamma R T^*}\,A_t = \frac{p^*}{\sqrt{RT^*}}\sqrt{\gamma}\,A_t$$

$$= p_0 \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}} \frac{1}{\sqrt{RT_{01}}} \left(\frac{2}{\gamma+1}\right)^{-\frac{1}{2}} \sqrt{\gamma}\,A_t$$

$$= \frac{p_0 A_t \sqrt{\gamma}}{\sqrt{RT_{01}}} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}} = 71.20 \text{ kg/s}$$

(b) $\frac{A_{exit}}{A_t} = 2 \quad \therefore M_{exit} = 2.20$

(c) $\frac{p_{exit}}{p_{01}} = \frac{1}{10.693} \quad \therefore p_{exit} = 28056 \text{ Pa}$

(d) For $\frac{A_{exit}}{A_t} = 2$, $1.0644 < \frac{p_{01}}{p_{exit}} < 10.693$

$$\therefore \frac{1}{10.693} < \frac{p_{exit}}{p_{01}} < \frac{1}{1.0644}$$

$$\therefore 28055 \text{ Pa} < p_{exit} < 281849 \text{ Pa}$$

3.5 Wind tunnel design

★★★ 25-30 mins

A blow-down wind tunnel is designed for a Mach number of $M = 3$. The reservoir pressure is $10\ \mathrm{bar}$.

(a) What will be the pressure in the test section? (Give 3 decimal places)

(b) What will be the pressure at entry to the diffuser, assuming a normal shock wave at the end of the test section? (Give 3 decimal places)

(c) A fixed area second throat is to be designed for the wind tunnel. What will be its minimum cross-sectional area, determined from flow starting considerations, if the test section has a cross section 8 cm x 4 cm? (Give 2 significant figures)

(d) What will be the Mach number at the second throat once the tunnel is running steadily? (Give 2 decimal places)

(a) $\frac{p_1}{p_0} = (1+\frac{\gamma-1}{2}M^2)^{-\frac{\gamma}{\gamma-1}} = 0.02722$

$\therefore p_1 = 0.272\ bar = 27223.684\ Pa$

(b) Normal shock wave at the end of the test section

$\frac{p_2}{p_1} = \frac{2\gamma M^2}{\gamma+1} - \frac{\gamma-1}{\gamma+1} = \frac{31}{3}$

$\therefore p_2 = 2.813\ bar = 281311.401\ Pa$

(c) $A_1 = 8cm \times 4cm = 32cm^2 = 3.2\times10^{-3}\ m^2$

$\because M_2^2 = \frac{2+(\gamma-1)M^2}{2\gamma M^2-(\gamma-1)} = \frac{7}{31} \quad \therefore M_2 = \sqrt{\frac{7}{31}}$

$\therefore (\frac{A_1}{A^*})^2 = \frac{1}{M_2^2}[\frac{2}{\gamma+1}(1+\frac{\gamma-1}{2}M_2^2)]^{\frac{\gamma+1}{\gamma-1}} = 1.9332$

$\therefore A^* = \frac{A_1}{\sqrt{1.9332}} = 2.3015\times10^{-3}\ m^2$

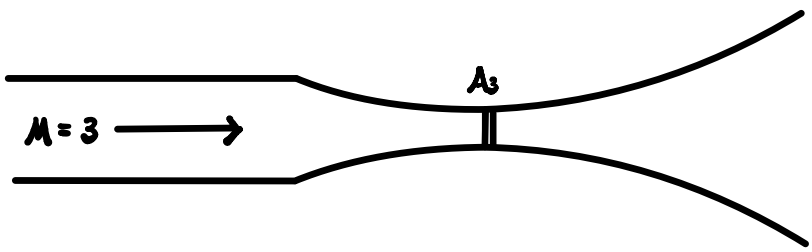
Let $M_D \doteq 0.99$,

$$\left(\frac{A_2}{A^*}\right)^2 = \frac{1}{M_D^2}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M_D^2\right)\right]^{\frac{\gamma+1}{\gamma-1}} \doteq 1.0002$$

$$\therefore A_2 = \sqrt{1.0002}\,A^* \doteq 2.3017\times10^{-3}\ m^2$$

(d) At steady state, to minimise the running power,

shock immediately after second throat



$$\therefore A_3 = 2.3017\times10^{-3}\ m^2$$

$$\therefore \frac{A_1}{A^*}\ (\text{for } M=3) = \frac{1}{M^2}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{\gamma-1}}$$

$$= 17.932$$

$$\therefore A^* = \frac{81}{343}A_1 = 7.557\times10^{-4}m^2$$

$$\therefore \frac{A_3}{A^*} = 3.046$$

$\therefore M_3 = 2.65$ according to the tables